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Alex Monito Nhancololo

Ph.D. student in Statistics and Probability

A. M. Nhancololo is a Ph.D. student in Statistics and Probability at the Institute of Mathematics, Statistics and Computer Science, University of São Paulo (IME–USP), advised by Prof. Airlane Pereira Alencar. He is currently (01/01 - 30/06, 2026) a visiting researcher at Jaume I University, advised by Prof. Jorge Mateu.

He holds an M.Sc. in Statistics and Agricultural Experimentation from the Federal University of Lavras (UFLA), advised by Prof. João Domingos Scalon, where he received the Best Thesis Award (2024). He earned his Bachelor of Education in Mathematics with a specialization in Statistics from Universidade Save, advised by Prof. André Silvestre Cuinica.

Additionally, he is pursuing an M.Sc. in Financial Engineering at WorldQuant University and an MBA in Artificial Intelligence and Big Data at the Institute of Mathematical and Computer Sciences, University of São Paulo (ICMC–USP), advised by Prof. Caetano T. Júnior. He was awarded third place in the Spatial AI Challenge 2024 (I-GUIDE), alongside Julian Huang and Yue Lin (University of Chicago).

His primary research interests lie in spatio-temporal statistics and their applications to the social sciences, environmental studies, ecology, agriculture, and medicine. Methodologically, his work focuses on geospatial intelligence (GeoAI), spatial point processes, lattice/area data, and geostatistics.

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[Honors & Awards](#)

2025 · I-GUIDE

Spatial AI Challenge 2024

Awarded third place in the Spatial AI Challenge 2024 (I-GUIDE), together with Julian Huang and Yue Lin (University of Chicago).



Spatial AI Challenge 2024

Fostering FAIR Data and Open Science Practices Using the I-GUIDE Platform

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Welcome to the Spatial AI Challenge 2024

The Spatial AI Challenge 2024 is an international initiative at the crossroads of geospatial science and artificial intelligence. Hosted on the cutting-edge I-GUIDE Platform, this challenge brings together researchers, data scientists, AI enthusiasts, and geospatial professionals to develop innovative, responsible, and reproducible solutions for some of today's most pressing sustainability challenges.

Winners of the Spatial AI Challenge 2024

We are proud to recognize the top teams whose submissions stood out for their innovation, technical rigor, and commitment to open science. After a competitive review process, the following projects have been selected as this year's winners:

🏆 Winner

[GeoMapCLIP](#)

Team: Purdue RCAC

Team members: Jungha Woo, Elham Jebalbarez Sarbijan — *Purdue Research Computing*

Summary:

This project developed *GeoMapCLIP*, a fine-tuned extension of GeoCLIP, to automatically localize unknown geospatial images using visual cues. Judges praised its well-structured, reproducible notebook, engaging storytelling, and generalizability across domains like agriculture, hydrology, and historical mapping.

🥈 Second Place

[Wildfire Threat Detection for Transportation Infrastructure using U-Net for Semantic Segmentation](#)

Team: CNA Corporation

Team members: John Crissman, Joel Diaz, Angie De Groot, Jeremiah Huggins, Lars Hanson, Steven Habicht, Matthew Prebble, Shaun Williams, Carey Whitehair-Conde, Rebekah Yang — *CNA Corporation*

Summary:

This submission presented a compelling approach to identifying wildfire risk for critical infrastructure using deep learning and open data. It leveraged a U-Net-based pipeline and performed well in data readiness, FAIR practices, and applicability to real-world emergency response planning.

🥉 Third Place

[Pipeline for Roof Material and Geometry Classification](#)

Team: Roof Mapper

Team members: Julian Huang, Yue Lin — *University of Chicago*; Alex Monito Nhamcololo — *University of São Paulo*

Summary:

Focusing on disaster resilience, this project used satellite imagery and OSM data to classify roof materials across urban environments. Reviewers appreciated its real-world applicability, integration of open tools, and clear documentation.

2025 · UFLA

2024 Best Master's Thesis Award in Statistics

Award for the Best Master's Thesis in Statistics, defended in 2024.

Advisor: Prof. João Domingos Scalón



2025 · WordQuant University

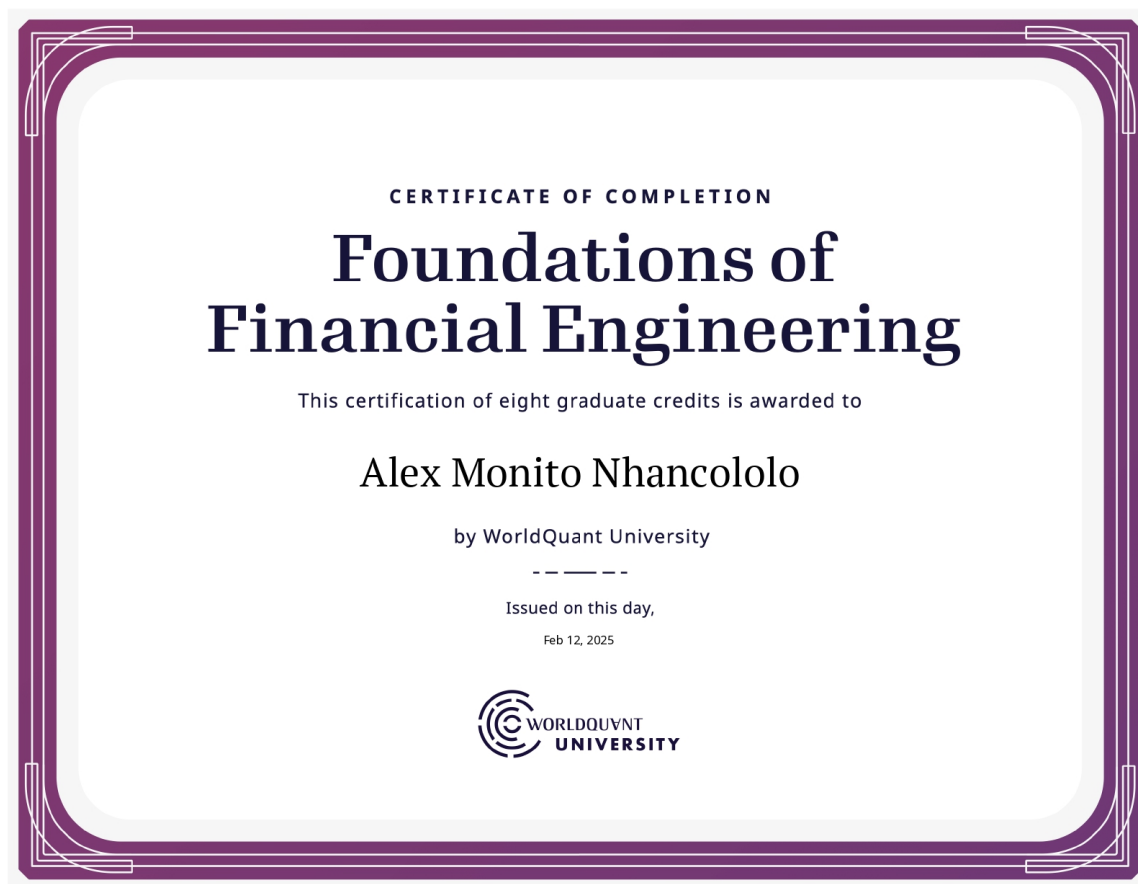
Foundations of Financial Engineering

Foundations of Financial Engineering (FFE) Certificate, distinction (80% average).

Applied linear algebra and Python for filtering, summarizing, and transforming structured and unstructured financial data

Prepared datasets for econometrics, machine learning, and deep learning models

Conducted quantitative analysis of financial risks including credit risk, volatility, and liquidity



2017: 04. Secondary School of Inharrime

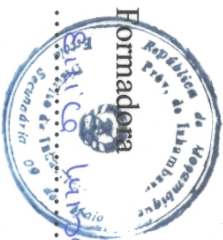
Poultry Production

Outstanding Student in the Poultry Production Course



REPÚBLICA DE MOÇAMBIQUE
GOVERNO DA PROVÍNCIA DE INHAMBANE
DISTRITO DE INHARRIME
ESCOLA SECUNDARIA 09 DE MAIO DE INHARRIME
DIPLOMA DE HONRA

*Este diploma constitui um reconhecimento ao aluno Alex Monito Nhancololo, pela
mérita no desempenho das actividades, durante o curso de avicultura, tendo se
destacado como o melhor estudante, com a classificação de 18 valores*



Director


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